

Auditory Materiality: Exploring Auditory Effects on Material Perception in Virtual Reality

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Figure 1: From left to right: (a) A participant scratching the prototype with his index finger while wearing a VR tracker. (b) The prototype is simulating earthenware with dynamic auditory feedback in VR.

ABSTRACT

This study examines the use of dynamic auditory cues to mimic the tactile sensation of touching materials in virtual reality (VR). A prototype employing finger pressure and movement input on a 3D-printed PLA board, a load cell, and eight piezoelectric sensors was developed. It emulates material textures like cardboard or earthenware through real-time audio feedback from pre-made audio recordings of scratching materials triggered by user interaction. A qualitative pilot study found that auditory feedback boosts immersion and material perception in VR. This suggests auditory cues are effective for enhancing material interactions in potential applications like virtual interior design, material selection, virtual museums, and gaming. Despite its limited sample size restricting generalizability, the study points to exciting future research avenues and enhancements in multimodal feedback systems.

Index Terms: Auditory feedback, virtual reality, material perception, auditory illusion, material simulation.

1 INTRODUCTION

Emerging technologies are developing rapidly, and it is now easier for people to enjoy virtual reality (VR) in their daily lives, with visual, haptic, and auditory feedback. VR opens up the potential for new ways of experiencing, with auditory feedback significantly influencing human perception; it can create material illusions to enhance immersion [1, 7]. While devices simulating virtual materials are advancing [11], research on sound-induced illusions in VR remains limited. Unlike complex haptic systems, auditory feedback offers a more practical approach to realism. Recent efforts focus on

leveraging sound-induced illusions using technologies like modal sound synthesis [14], data-driven sound modeling [8], and haptic-sound coupling [3]. These previous studies demonstrate the importance of auditory feedback in creating immersive VR experiences.

Simulating realistic materials in VR involves two main approaches: physical haptic feedback using devices such as force feedback and surface texture simulation [13], and pseudo-haptic feedback through visual or auditory cues [15]. While physical feedback is effective, its limited range of materials and device constraints can reduce immersion. In contrast, cross-modal pseudo-haptics offer greater flexibility, with auditory feedback showing potential. However, research focusing solely on sound-based pseudo-haptics remains rare, leaving room for exploration in simulating material textures using auditory cues.

In this paper, we suggest an auditory feedback system that outputs different dynamic sounds made of recorded audio samples of a finger scratching material surfaces. The sample playback is controlled in real-time based on the finger pressure and finger movement measured on a physical 3D-printed PLA board. We developed a prototype using 1 load cell and 8 piezoelectric sensors to capture the user's finger pressure and movement. The 3D-printed PLA board offers the user only a single tactile texture, and the goal of this research is to create the illusion of different other materials by providing dynamic auditory cues to the user while they are always scratching the same physical PLA board. Therefore, our system improves the user's perceptual possibilities of the texture surface of the single PLA board upon touch, effectively mimicking the sensations associated with touching and scratching different materials, such as cardboard and pottery.

Within the domain of using auditory feedback for the creation of material illusions, there are also several possible avenues: in general, recorded or synthesized sound can be used, and within synthesized sound, several synthesis techniques are available [2]. Using sound synthesis, such as in Chan, Tymms and Colonnese's work [2], certainly allows for greater flexibility in dynamically changing the sound's parameters in real time and aligning them with a dynamic input. However, in this first prototype and first pilot study, we used

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recorded audio samples in order to efficiently have more accurate, realistic-sounding sounds of a selection of scratching material surfaces at hand.

This work presents the following contributions: (1) The design of the proposed hardware system: a prototype based on a 3D-printed PLA board, which includes finger pressure measurement and finger contact detection. (2) A software system consisting of an algorithm to dynamically trigger pre-recorded audio files and adjust them in their amplitude and length based on the measured input from the hardware system, so as to enhance immersion in a virtual reality setting. (3) A pilot study was conducted to evaluate the effectiveness of the system, comparing two conditions with and without the hardware part of the prototype. (4) Potential VR applications of the system were laid out, such as interior design and virtual museums, and directions for future research were determined.

2 METHODS

We have developed a unique pressure-based auditory feedback system designed to output dynamic sound cues in real-time based on the user's finger pressure input and finger movement. As a part of this system, we designed a software algorithm that triggers and dynamically controls pre-recorded audio samples that we recorded by ourselves, scratching different material boards with the finger in the recording studio. This system enhances the immersion of material perception, enriching the user's experience in VR. An overview of our approach is shown in Fig. 2.

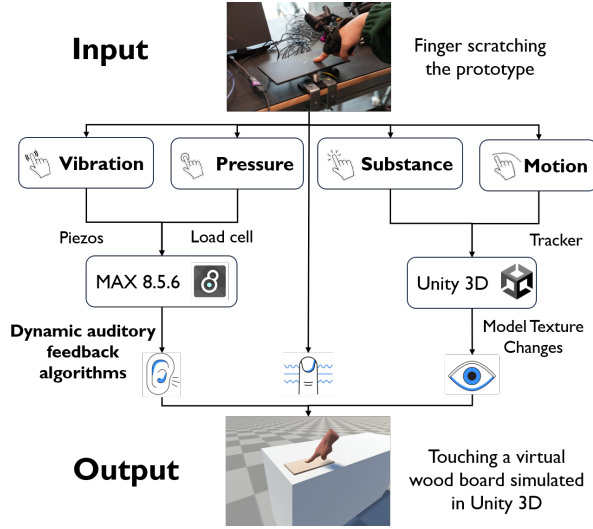


Figure 2: System framework diagram of the prototype using 1 load cell and 8 piezoelectric sensors to capture finger pressure and movement input and provide dynamic auditory feedback through Max 8.5.6. A 3D-printed PLA board measures the user's finger pressure and movement, and the Unity 3D scene provides visual models of different simulated material textures.

2.1 Hardware Design

The prototype consists of a 3D-printed PLA board, 8 piezoelectric sensors, and 1 load cell. The load cell is mounted under the board using bolts, with a clamp at the base to secure the board firmly to a table. The 8 piezoelectric sensors are attached underneath the board using a soft adhesive. The load cells measure the pressure when the user's fingers touches the board. At the same time, the piezoelectric sensors capture the the amplitude of the friction generated by the user moving their finger, as is shown in Fig. 3

Finger-Touch Interface: We used a 3D-printed PLA board as an interface to provide a haptic non-zero texture sensation to the

user touching it. We chose polylactic acid (PLA)[10, 12], the most used material in 3D printing, as the material for this board.

An Arduino UNO board is used to manage sensor data collection and send it via serial communication to an HP OMEN 15 laptop running Max 8.5.6. Similarly, signals captured by the piezoelectric sensors are transmitted to Max 8.5.6 through a MOTU 828ES audio interface connected to the same laptop. The load cell model is an OLUCA AT8501 with a range of 1kg and an accuracy of 0.1g.

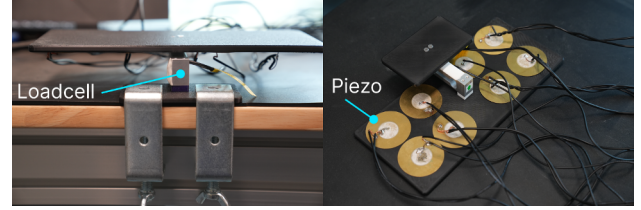


Figure 3: From left to right: (a) A load cell (OLUCA AT8501 1kg) is bolted underneath the 3D-printed PLA board, and 2 clamps are used to secure the entire prototype to the table. (b) Upside-down view of prototype, showing one load cell and 8 piezoelectric sensors installed under the PLA board to detect the pressure and movement of the finger scratching.

2.2 Sample Library & Software Design

This research proposes a dynamic auditory feedback control method, which is primarily realized in software, based on input from the user's finger pressure and finger movement. The primary goal is to simulate the auditory interaction of fingers scratching surface textures of different materials in virtual environments, while in reality they touch always the same PLA board. When there is a noticeable scratching sound, we observed that the heavier the finger pressure when scratching a material's surface, the faster the finger accelerates, and the louder the auditory feedback. However, the sound quality (timbre) of the same material surface will be different when scratched with little or strong force, not only the amplitude. Therefore, we used two audio files of each material that we recorded by ourselves under laboratory conditions, under applying soft and heavy force, to produce a sample library that was later used to simulate the material surface textures. We control the audio recording's playback (start and end times) dynamically according to the user's finger pressure and finger movement, and we also apply dynamic amplitude control to the playing sound file, so to simulate different forces of scratching the virtual materials. The dynamic amplitude control function of the two available audio recordings for heavy and light pressure is written as a piecewise function. By using this function to dynamically play back the sound of scratching a material surface, we can smoothly transition in real time between the two available audio recordings. The general piecewise function for the volume $V(F)$ can be expressed as:

$$V(F) = \begin{cases} 0, & F < F_{\min}, \\ \sum_{i=1}^n (k_i F + b_i), & F_{\min} \leq F < F_{\max}, \\ V(F_{\max}), & F \geq F_{\max}, \end{cases}$$

Where:

F is the input pressure or weight (grams, g);

$F_{\min}, F_{\max}$ are the boundary points of the piecewise function;

k_i, b_i are the coefficients of each linear segment.

2.3 Material Selection

Materials to simulate: We selected four different materials that we simulated with auditory feedback in this research. This selection is

based on what previous research has presented as more efficient in recognizing materiality through auditory cues, or as more common for users to expect in contexts such as museum artifacts or everyday listening settings, and therefore with greater application potential [9, 8, 2, 6, 4, 5]. We selected four materials from the material library that have great application potential and can effectively be simulated with our prototype, namely plywood, earthenware, ABS (Acrylonitrile Butadiene Styrene), and cardboard. The four selected materials are shown in Fig. 4.

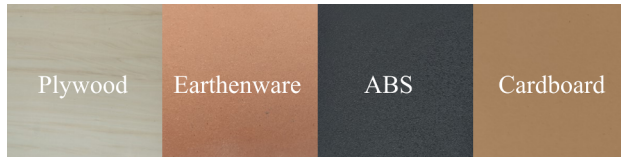


Figure 4: The four materials selected to be simulated in our hardware/software system.

3 EVALUATION

We conducted a qualitative pilot study to explore the potential of using dynamic auditory feedback to simulate the sensation of scratching different material surface textures in VR. The goal was to understand whether dynamic auditory feedback, while the user is touching a non-zero texture of the 3D-printed PLA board as the finger-pressure interface, could create a convincing illusion of different material surface textures. This study aimed to gather insights into the immersive experience and perception of materials through auditory feedback and explore how the hardware/software system might influence material interactions in virtual environments.

3.1 Equipment and Setup

Participants were placed in an immersive virtual environment rendered through a HTC VIVE Pro 2 head-mounted display (HMD). They wore a tracker on their dominant hand and were tracked with two HTC VIVE locators. To ensure an optimal auditory experience, participants used Pioneer DJ HDJ-CUE1 headphones (uncalibrated for this pilot study), chosen for their closed-back design to reduce external noise and provide a linear frequency response. The prototype, which simulates the material sensation, was placed on the table in front of the participant. Based on finger pressure and movement, the dynamic auditory feedback was played through the headphones. The study was conducted in a quiet room to minimize distractions. The setup is shown in Fig. 5.

3.2 Study Design and Procedure

The research question was evaluated with a comparative trial comparing two conditions: (1) users only seeing a virtual material board in the HMD and “touching” it in the air through a worn hand tracker vs. (2) users seeing the virtual material board in the HMD while touching with their finger a 3D-printed PLA board. Each participant needed to complete 4 materials $\times$ 2 conditions = 8 trials. The participants could freely scratch the prototype under each condition and listen to the sound for 30 seconds. In the no-PLA-board condition, participants do not receive the dynamic auditory feedback, but instead hear only the same unprocessed and uncontrolled audio recording upon touching the virtual board “in the air” with the hand-tracker. The sound plays when their finger scratches virtual materials and stops when they move away from the virtual board. The visual parameters remain exactly the same for both conditions. In addition, half of the participants started testing with condition (1), and the other half started testing with condition (2), in a counterbalanced manner.

In this pilot study, participants explored the immersive experience of scratching different virtual materials with and without a



Figure 5: Setup: A participant is scratching the prototype using his tracker-wearing hand and wearing the HMD and headphones on his head, while looking at the virtual material board and listening to the auditory feedback of scratching its surface texture.

touchable, 3D-printed PLA board. Participants interacted with four materials (plywood, earthenware, ABS, and cardboard) under two conditions: one with the PLA board, which provided them with dynamic auditory feedback while their finger touched the non-zero haptic texture of the board, and the other without the PLA board, where only visual feedback was provided in the VR environment as users were moving their finger in the air while wearing a hand tracker. Before the start of the experiment, the gain settings on the sound card controlling the headphone output volume were calibrated to the same parameters for all participants to ensure consistent auditory feedback amplitude across all trials. Each participant could freely explore the materials under both conditions.

The software used for the pilot study randomly selected one of the four selected materials for each trial, and displayed it on a virtual table in front of the participant. Participants were instructed to interact with the materials and listen to auditory feedback for 30 seconds per material. After all trials were completed, a 10-minute semi-structured interview was conducted to gather qualitative feedback on the participant’s experience. The session took approximately 30 minutes for each participant, with participants allowed to pause and rest at any time during the pilot study.

The semi-structured interview included the following questions to gain insights into the participants’ experience:

1. Which material is the most authentic?
2. Which of the two conditions do you think is better?
3. What application potential do you think our prototype has?
4. Would you like to use our prototype in the future?
5. Do you think the dynamic sound feedback meets your expectations?
6. Do you think the dynamic sound feedback is caused by the changes in your finger movement?
7. Does the dynamic auditory feedback remind you of the materials being displayed?

3.3 Participants

In this study, we recruited 10 participants (6 self-identified as female and 4 self-identified as male) between the ages of 20-28 ($M = 22.8$, $SD = 2.1$), and all participants’ dominant hand was the right

hand. All participants had normal hearing; 7 had experienced VR before, and 1 had known before the research domain of auditory illusions. All participants were compensated with a locally equivalent amount of 10 USD for their participation. The study was approved by our Institutional Review Board (IRB).

4 RESULTS AND DISCUSSION

Eight people subjectively felt that the sense of presence was better in the condition with the PLA board. The dynamic auditory feedback combined with the physicality of the haptic non-zero surface texture sensation provided to them through the PLA board made the scratching action more plausible to them. The multisensory cues helped users perceive material properties such as texture, roughness, and hardness. However, some participants mentioned that when the PLA board did not visually align with the simulated material in the VR scene, it caused some confusion [P3, P7]. This suggests that while combining sensory cues can enhance immersion, mismatched sensory feedback can disrupt the experience.

Participants identified numerous possible applications for the prototype within VR contexts, including virtual interior design, museums, material selection (notably in architecture), and gaming. Informal feedback corroborates these uses, underscoring the system's capability to enhance immersion and improve interactions in virtual environments requiring material assessment or manipulation.

The pilot study's limited sample size and qualitative nature restrict its generalizability and statistical reliability. Future research should include more diverse groups and quantitative methods to validate findings. Expanding materials and interaction types can enhance understanding. Additionally, testing varied surface textures in VR and exploring sound synthesis for real-time adaptability is suggested. Assessing the hardware prototype in diverse VR applications could show its versatility. Investigating long-term effects of multimodal feedback on user engagement and learning could offer insights for improving VR interactions. These directions will strengthen our findings and aid the development of immersive virtual environments.

5 CONCLUSION

This research explored the potential of dynamic auditory feedback for simulating material surface textures in virtual reality (VR). The hardware/software system that we developed shows how real-time auditory feedback can enhance the perception of materials in VR by matching auditory cues to the users' interactions, providing them with an immersive and engaging experience. Based on their finger pressure and finger movement applied to a material surface texture, participants hear auditory feedback associated with four selected materials: plywood, earthenware, ABS (Acrylonitrile Butadiene Styrene), and cardboard.

The study shows that adding dynamic auditory feedback to a 3D-printed PLA board's tactile texture improved participants' ability to identify four virtual materials. Participants found it more immersive compared to using only a virtual board with less dynamic audio feedback. This underscores sound's impact on VR material perception, even without dynamic haptics. Although the small sample size and qualitative approach limit broad conclusions, this pilot suggests value for educational and design applications. Future work should involve larger, diverse groups and rigorous quantitative methods to enhance and validate the prototype's VR use.

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